Worksheet & Answer Key

When resistors are connected in parallel, each resistor experiences the full voltage of the circuit. The combined resistance of a system of resistors in parallel is less than the individual resistances would be if they were connected in series. This occurs because each resistor carries only a portion of the total current, and the total current is greater than it would be for each resistor alone. The total resistance of the system is determined by the reciprocal of the sum of the reciprocals of the individual resistances.

Word Bank:

series, voltage, current, portion, reciprocal, total, resistors, system

Fill in the blanks:

1. Circuits with resistors connected in _____ have a combined resistance less than the sum of the individual resistances.
2. The full _____ is experienced by each resistor when they are connected in parallel.
3. A _____ of the total current is carried by each resistor in a parallel circuit.
4. The _____ of the system is determined by the reciprocal of the sum of the reciprocals of the individual resistances.
5. In a parallel circuit, the total _____ is greater than it would be for each resistor alone.
6. Each resistor experiences only a _____ of the total current.
7. A _____ circuit has a combined resistance that is equal to the sum of the individual resistances.
8. The _____ of the system is less than the sum of the individual resistances when resistors are connected in parallel.
9. Circuits with resistors connected in _____ have a greater total _____.

■ Answer Key (for teachers only):

1. series
2. voltage
3. portion
4. reciprocal
5. total
6. resistors
7. system
8. parallel